

Detection of Fake News in Nepali Language Using Transformer-Based Natural Language Processing Models

Bishal Mahjarjan ^a, Anup Shrestha ^b

^{a, b} Department of Electronics and Computer Engineering, Thapathali Campus, IOE, Tribhuvan University, Nepal

✉ ^a bishalmhj68@gmail.com, ^b anupluckystha@gmail.com

Abstract

The spread of fake news is a severe problem in Nepal, particularly on social media. It is further complicated by the fact that Nepal uses 'Nepanglish', a mixed language that standard detection mechanisms can't deal with. That is why we propose applying a new approach to the detection of false news in Nepali based on a state-of-the-art language model called mBert. A major issue is that there is not much Nepali language data available for training such models. We avoid this by using advanced methods of creating additional training data. We will train our model on the increased data and test on standard performance metrics. We aim to build a highly accurate and efficient Nepali fake news detector. Apart from this specific application, our study indicates that robust language models can prove to be very useful to under-represented languages with minimal digital footprints, providing an effective tool for other under-represented languages. The research will help in realizing the spread of disinformation in Nepali social media and provide an effective tool for media organizations, educators, fact-checkers, and policy-makers to combat the spread of fake news and improve social relationships in Nepal.

Keywords

Low-resource, mBERT, NLP, Nepanglish, Transformer

1. Introduction

The general spread of social media networks such as Facebook, Twitter, and TikTok has turned fabricated news into a fast-emerging concern in Nepal [1]. Nepali situations' misinformation tends to rely on sociopolitical sensitivities for instilling disturbance, bifurcating social opinions, and discrediting reliable news outlets. Widespread linguistic variability, dominance of local dialects, and paucity of annotated datasets crafted particularly for the Nepali language make it more challenging to detect and counteract fake news in Nepal [2]. Recently, using most state-of-the-art transformer-based models such as BERT, mBERT, and XLM-RoBERTa, the performance of fake news detection has noticeably improved in the world [3][4].

These models use the attention mechanism, which is capable of capturing fine-grained contextual relationships of text and performs classification tasks very well [5]. In fact, the transformer-based approaches have surpassed the traditional machine learning models at an accuracy rate of up to 98 percent while showcasing the state-of-the-art performance in Arabic and Spanish languages [6]. Although these works succeed in most of the languages, these works fall short in their utilization in the Nepali Language, and so a deadly deficiency arises in confronting mis-informing information. Specifically, Nepal has a very unique challenge when distinguishing fake news because it has a multilingual society, and most of the citizens mix English with Nepali, or so-called "Nepanglish" [7]. Such linguistic diversity is a challenge for traditional models to achieve high accuracy. Moreover, with over 16 million Facebook users and a substantial number of TikTok users in Nepal as of 2024, the rapid spread of fake news through social media has become a serious concern.[8] Nepalese social media users themselves suggest that they

encounter fake news on a daily basis at about 38 percent, which reflects a growing demand for a good detection system [9]. This research tries to solve these issues by utilizing transformer-based models fine-tuned for Nepali to build a robust fake news detection system. Different domain-specific fine-tuning techniques, data augmentation, and mixed language corpora will be utilized to improve model performance. Publicly available datasets will be augmented with Nepali-specific news articles and synthetic data to train and test the models for scalability and generalizability.

This will not only fill in the gap within the critical literature of identifying hoax news in Nepali language but also function as a sorely needed device within the perpetuation of social concord and facilitation of establishing well-education-based decision-making in the country. This treatise will then become the catalyst on which further people can use extended cutting-edge NLP approaches to additional low-represented languages-a building block to technology advancement inclusion within the region.

The motivation in addressing fake news in Nepali arises with increased misinformation, which might result in the wide misinterpretation of news, damaging social harmony and hurting trust in reliable sources. One thus requires a robust system for the analysis and classification of fake news in Nepali, which poses additional challenges due to its linguistic complexity and mixed scripts such as Nepanglish [7]. Transformer-based models, such as mBERT have achieved F1 scores of over 94 percent with fine-tuning on this task with great promise in this respect [6]. Such models are good at capturing the nuances of the Nepali language, while back-translation and paraphrasing techniques [10] can be used to create larger datasets and therefore overcome the scarcity of annotated corpora. This enhances the generalizability of models across text types and domains,

making them resilient to shifting linguistic patterns and better suited to combat the spread of misinformation.

2. Related Works

Numerous recent studies have showcased significant advancements in fake news detection using deep learning and NLP techniques. Transformer-based models, in particular, have demonstrated superior performance in text classification tasks due to their ability to capture contextual relationships effectively. These models have outperformed traditional machine learning and neural network-based approaches, addressing challenges such as data sparsity and linguistic diversity.

For example, [11] (2021) explored the application of pretrained transformer-based models for detecting COVID-19-related fake news. By employing custom loss functions and contextual embeddings, the study achieved high weighted average F1-scores, highlighting the effectiveness of domain-specific fine-tuning [1]. Al-Yahya et al. (2021) conducted a comparative study of neural network and transformer-based models for Arabic fake news detection, reporting an increase in F1-scores from 0.83 using GRU to 0.95 with QARiB, a transformer-based model, indicating their superior performance [2]. Sánchez Gálvez et al. (2024) proposed a fake news detection system for Spanish using transformer models such as BETO and mBERT. The study achieved an accuracy of 98 percent, demonstrating the robustness of transformers in handling linguistic diversity [12].

Glazkova:2023[13] (2023) introduced a novel data augmentation method combining Seq2Seq and Natural Language Inference (NLI) models to enhance training data quality for fake news detection. This approach improved generalization and retained class labels effectively, resulting in higher F1-scores and AUC values across multiple datasets . Azizah et al. (2023) analyzed the performance of transformer models such as BERT, ALBERT, and RoBERTa for detecting fake news in Indonesian. The study found ALBERT to be the most efficient, achieving 87.6 percent accuracy with reduced computational requirements .

Al-Yahya et al. (2021) [5] investigated fake news detection in the Arabic language, using a dataset containing Arabic fake news examples. They compared the performance of GRU-based and transformer-based models, specifically the QARiB model, in a GPU environment. The study reported a high F1-score of 0.95, demonstrating the model's excellent ability to correctly classify fake news while minimizing false positives and negatives. The researchers highlighted the model's superior contextual understanding, which is crucial for handling the complex nature of the Arabic language. However, the study faced limitations due to the small size of the dataset, which may impact the generalizability of the results to larger, more diverse collections of Arabic fake news.

Sánchez Gálvez et al. (2024) [6] conducted their research on detecting fake news in Spanish, specifically targeting misinformation spread on Twitter. This study utilized GPU-powered resources and the PyTorch framework to implement mBERT and BETO models. The model achieved an

impressive 98 percent accuracy, underlining its high efficacy in accurately identifying fake news across different linguistic forms within Spanish. The study's strength lies in its consideration of linguistic diversity within the Spanish language, making the model robust across variations. However, its applicability is limited because the study focused exclusively on Spanish, and its findings may not be easily transferable to other languages.

Shakya and Poudyal [14] explores advanced methodologies for detecting fake news using deep learning models. It employs the SBFN dataset to evaluate the performance of various architectures, including Long Short-Term Memory (LSTM), BERT, and hybrid approaches combining different model types. The study demonstrates how these models can effectively identify patterns in textual data to differentiate between fake and real news, addressing challenges unique to low-resource settings. This research highlights the potential of deep learning in combating misinformation while emphasizing the importance of robust datasets and preprocessing for achieving high accuracy in multilingual environments.

Azizah et al. (2023) [12] worked with an Indonesian fake news dataset and utilized GPU-powered resources alongside TensorFlow for training transformer-based models, such as BERT, ALBERT, and RoBERTa. Their models achieved an accuracy of 87.6 percent, demonstrating an efficient approach even in resource-constrained environments. This suggests that their model can effectively handle real-world challenges where computing power is limited. Despite its advantages, one weakness identified in this study is the slightly lower accuracy when compared to larger, more resource-demanding models, which might hinder its scalability to larger datasets.

These studies highlight diverse approaches to fake news detection across languages and datasets, each with unique strengths such as high accuracy, domain-specific adaptation, or efficient performance in resource-constrained settings. However, they also share limitations, such as narrow dataset focus, dependency on language-specific models, and lower scalability in certain contexts.

3. Methodology

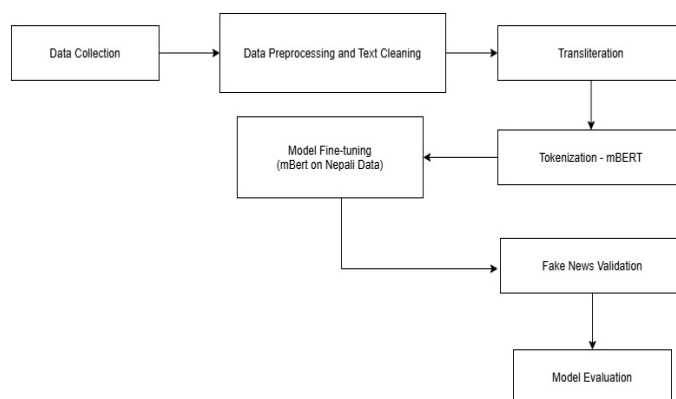


Figure 1: System Block Diagram

3.1 Data Collection

Data gathering is the foundation for the fake news detection pipeline. Here, a dataset of Nepali news articles is gathered with labels as either "real" or "fake," comprising 30,000 fake and 40,000 real data points, totaling 70,000 items. During data gathering, multiple sources are usually utilized to ensure an exhaustive and accurate dataset. Real data are primarily collected through web scraping from various reputable websites, including prominent Nepali news portals like Kantipur, Setopati, and Annapurna Post, as well as other news agencies. These sources are generally highly respected for their commitment to editorial scrutiny and fact-checking, thus providing reliable information.

Fake news data are collected from several sources, including social media platforms (e.g., Facebook, TikTok), unreliable blogs, and through data augmentation techniques such as back translation and the use of AI tools. Additionally, some fake data points are manually added to ensure a comprehensive representation of misinformation. Another source that included in data collection is parody social media news pages. These pages intentionally create satirical or humorous content, often mimicking the style of real news outlets. While entertaining, it's crucial to acknowledge that data from these pages should not be treated as factual. However, they can serve specific research purposes, such as studying humor or analyzing public perception of news within the context of social media. To ensure effective model training, the dataset are diverse, covering topics like politics, health, culture, and economics.

3.2 Data Preprocessing and Text Cleaning

Raw text data often contains noise, which needs to be cleaned and standardized. Data preprocessing and text cleaning for Nepali text are also very much similar to other languages but it requires to be done with utmost attention to its unique linguistic features. While standard data cleaning practices like handling missing values, duplicates, and inconsistencies are applied, text cleaning for Nepali must pay special attention to Unicode encoding, which is crucial in properly representing Nepali characters. Punctuation removal, and number removal are generally useful, but stop word removal requires a Nepali stop word list. The rich morphology in word construction makes tokenization harder, typically requiring modified or bespoke tokenizers. Stemming and lemmatization are even more challenging due to the language's morphology, with very few easily obtainable tools available, necessitating searching research articles and potentially custom solutions.

In addition, text normalization ensures that words like (to increase) and (to grow) are handled consistently. This process helps eliminate noise, allowing the model to focus on meaningful information.

- **Raw text:** "नेपालको GDP वृद्धि ७% ले **बढ्न सक्छ।**"
- **Cleaned text:** "नेपालको GDP वृद्धि ७ ले बढ्न सक्छ।"

Figure 2: Data Preprocessing and Text Cleaning

3.3 Transliteration

Many users, especially on social media, write Nepali using Romanized script (e.g., "Nepali using English alphabets"). Transliteration is the process of converting this Romanized Nepali into the Devanagari script to maintain consistency. Nepali text often appears in both Devanagari script and Romanized form. Transliteration converts Romanized Nepali into Devanagari to ensure consistency. This is important because mixed-script data can confuse tokenizers and models.

- **Romanized text:** "Yo desh ko samasya dirghakalin xa."
- **Transliterated text:** "यो देशको समस्या दीर्घकालिन छ।"

Figure 3: Transliteration

3.4 Tokenization - mBERT

Tokenization involves splitting text into smaller units (tokens) that can be understood by machine learning models. The mBERT tokenizer uses subword tokenization, which breaks down words into smaller components, allowing it to handle rare or unknown words. This is particularly useful for a morphologically rich language like Nepali, where words often consist of roots, prefixes, and suffixes. Each token is then matched to mBERT's vocabulary. If the token is not present in the vocabulary, the tokenizer decomposes it into smaller subwords or characters. WordPiece builds the token by maximizing the number of matched characters starting from the beginning of the word.

- **Input text:** "नेपालको आर्थिक स्थिति दृढ छ।"
- **Tokenized output:** ["ने", "पाल", "को", "आ", "र्थिक", "स्थिति", "दृढ", "छ", "।"]

Figure 4: Tokenization

3.5 Model (mBERT on Nepali Data)

The multilingual BERT (mBERT) model is pre-trained on multiple languages, including Nepali. In the case of mBERT, this input can be in any of the 104 languages it supports. mBERT uses a Transformer encoder model to classify cross-lingual data. Text is encoded into numerical embeddings, and positional encodings are added to maintain word order. Multi-head attention allows the model to weigh the relative significance of words in relation to each other, capturing context across languages. Residual connections and layer normalization stabilize training. Every word representation is passed through a feed-forward network. This entire encoder block is stacked multiple times to create a hierarchical representation. In classification, the final representation of a special [CLS] token is used as a summary of the input and fed through a classification layer to produce probabilities for different classes. However, it needs fine-tuning to specialize in fake news detection for Nepali. Fine-tuning involves:

- **Input preparation:** Feeding cleaned, tokenized text into mBERT.
- **Task-specific training:** Training mBERT on the labeled dataset (real vs. fake news).
- **Adjusting weights:** The model updates its parameters based on the task.

Fine-tuning involves training the mBERT model on a labeled dataset of 70k Nepali news articles to detect patterns that distinguish real from fake news. The model uses contextual embeddings, meaning it understands the relationship between words in a sentence. For example, (Nepali) paired with (population) might indicate a demographic topic, whereas paired with (anthem) might indicate a cultural topic. Fine-tuning allows the model to specialize in Nepali-specific language and context. This process, especially with a dataset of 70,000 articles, typically requires significant computational resources, ideally including a GPU with at least 12GB of VRAM to accelerate the training. Additionally, a multi-core CPU with sufficient RAM is used. The training is commonly performed using deep learning frameworks like PyTorch with the Hugging Face Transformers library.

3.6 Evaluation

Verification and validation are the stages of evaluating the model's performance. If the model performs successfully, it may be accepted for usage in the intended real-world scenario. This paper approaches detecting false news as a binary classification problem. The proposed model relies on assessing Machine and Deep Learning methods. Performance evaluation is based on five metrics: accuracy, precision, F1-score[15]. Accuracy is the ratio of correct predictions to total input samples, with TP, TN, FP, and FN being True Positive, True Negative, False Positive, and False Negative, respectively.

Here's a breakdown of these terms in the context of false news detection:

- **True Positive (TP):** The model correctly identifies a piece of news as false.
- **True Negative (TN):** The model correctly identifies a piece of news as true.
- **False Positive (FP):** The model incorrectly identifies a piece of news as false when it is actually true (also known as a Type I error).
- **False Negative (FN):** The model incorrectly identifies a piece of news as true when it is actually false (also known as a Type II error).
- **N:** The total number of news articles in the dataset.

3.6.1 Precision

Precision is calculated as:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (1)$$

Out of all the news articles the model predicted as false, what proportion was actually false? A high precision score indicates that when the model says a piece of news is false, it is very likely to be true.

3.6.2 Recall

Recall is given by:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (2)$$

Out of all the news articles that are actually false, what proportion did the model correctly identify? A high recall score indicates that the model is good at finding most of the false news articles.

3.6.3 Accuracy

Accuracy measures the ratio of correct predictions to the total number of data samples:

$$\text{Accuracy} = \frac{TP + TN}{N} \quad (3)$$

3.6.4 F1 Score

The F1 Score is the harmonic mean of Precision and Recall:

$$\text{F1 Score} = 2 \cdot \frac{\text{Recall} \cdot \text{Precision}}{\text{Recall} + \text{Precision}} \quad (4)$$

This metric provides a balanced measure of both precision and recall. It is useful when there is an uneven distribution of classes (e.g., more true news than false news, or vice versa). A high F1 score indicates that the model has a good balance of both precision and recall.

4. Experiments and Result

This section details the training process for the fake news detection system using mBERT (multilingual Bidirectional Encoder Representations from Transformers). The system's objective is to classify news articles as either "fake" or "real," minimizing classification errors. System performance will be evaluated using standard metrics for classification tasks.

A pre-trained mBERT model will be fine-tuned on a labeled dataset of news articles. The dataset is partitioned into training, validation, and test sets to ensure robust model development and evaluation.

4.1 Dataset Composition

Table 1 presents the composition of the dataset used to train the mBERT model.

Table 1: Dataset Composition for Training mBERT

Category	Number of News Articles
Fake	30,000
Real	40,000
Total	70,000

The dataset is split into training, validation, and test sets using stratified sampling to maintain the approximate ratio of fake to

real news articles in each subset. An 80/10/10 split is employed, resulting in the approximate sizes shown in Table 2.

Table 2: Dataset Split

Category	Number of News Articles
Training Set	57000
Validation Set	6500
Test Set	6500

4.2 Model Output

This section presents and analyzes the output of the fake news detection model on a set of news articles. Table displays the model’s predictions alongside the ground truth labels.

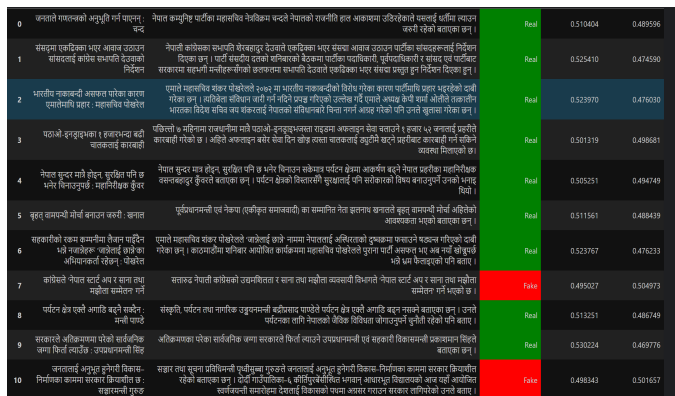


Figure 5: Model Output

This table presents the output of a fake news detection model. Each row corresponds to a news article, and the columns provide the following information:

- **Title:** The title of the news article.
- **Summary:** A brief summary of the news article's content.
- **Label:** The ground truth label indicating whether the news article is “Real” or “Fake”. This represents the actual known classification of the article.
- **Real Probability:** The probability assigned by the model to the news article being “Real”. This is a numerical value between 0 and 1.
- **Fake Probability:** The probability assigned by the model to the news article being “Fake”. This is also a numerical value between 0 and 1.

4.3 Model Performance

The table 3 presents the performance metrics for the Multilingual BERT model in classifying news as "Fake" or "True". The precision for identifying "Fake" news is 0.9445, while for "True" news it is 0.9181, resulting in an average precision of 0.9313. The recall, indicating the model's ability to find all instances of each class, is 0.9214 for "Fake" news and 0.9420 for "True" news, with an average recall of 0.9317. The F1 Score, a balanced measure, is 0.9329 for "Fake" and 0.9299

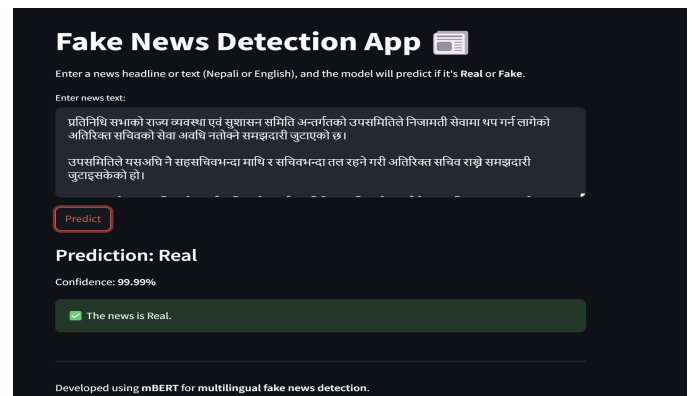


Figure 6: Output

Table 3: Performance Metrics for Multilingual BERT

	Fake	True	Average
Precision	0.9445	0.9181	0.9313
Recall	0.9214	0.9420	0.9317
F1 Score	0.9329	0.9299	0.9314
Accuracy			0.9314

for "True", averaging at 0.9314. The overall accuracy of the Multilingual BERT model on the dataset is 0.9314.

Additionally, fig 7 the performance of the Multilingual mBERT model, displaying the counts of correct and incorrect classifications: 3096 True Positives (correctly predicted Fake), 264 False Negatives (actually Fake, predicted True), 182 False Positives (actually True, predicted Fake), and 2958 True Negatives (correctly predicted True). This provides a detailed view of the model's classification results.

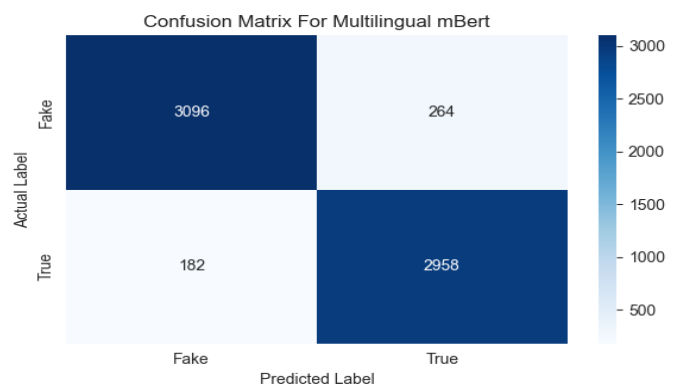


Figure 7: Confusion Matrix

4.4 Comparative Analysis

Fake news detection has traditionally been performed by [16] using machine learning algorithms such as Naïve Bayes, Support Vector Machine (SVM), and Logistic Regression. The performance of these traditional methods has been evaluated based on four key metrics: Accuracy, Precision, Recall, and F1-score. The results, as shown in the bar graph 8

However, mBERT (Multilingual BERT) has significantly advanced fake news detection by leveraging deep learning and context-aware representations. Unlike traditional models, mBERT is pre-trained on a vast amount of multilingual text using the transformer architecture, enabling it to capture deep

semantic meanings and contextual relationships. The results demonstrate that mBERT consistently surpasses traditional machine learning methods across all evaluation metrics, achieving higher Accuracy (93.14), Precision (93.13), Recall (93.17), and F1-score (93.14). This improvement is attributed to BERT's ability to learn contextual embeddings and its transfer learning capability.

By replacing traditional machine learning models with mBERT, fake news detection systems can achieve more robust, accurate, and scalable performance across multiple languages and domains.

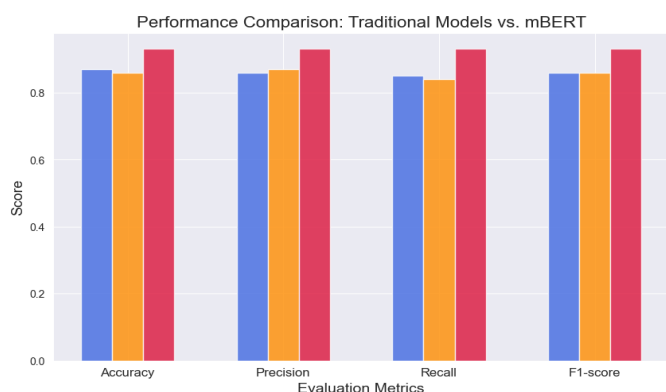


Figure 8: Comparative Analysis

5. Conclusion

This research has been able to demonstrate the effectiveness of mBERT in detecting fake news, especially in the case of low-resource languages like Nepali. By fine-tuning mBERT over a large dataset and employing data augmentation techniques, the model has been successful in showing promising results in properly detecting fake news articles. This work has referenced the potential of mBERT as a strong instrument in the battle against misinformation and towards an informed society, particularly in regions with few language resources.

6. Future Works

To further enhance the performance of mBERT-based fake news detection models, several promising avenues for future work can be explored. One significant one is expanding training dataset size and diversity, especially for low-resource languages, through methods like crowdsourcing, web scraping, and cooperation with news organizations. Looking further into more advanced data augmentation techniques such as generative adversarial networks (GANs) or contextualized word embeddings can be used to produce synthetic data that would improve model performance further. Mixing mBERT with other NLP models or machine learning models in an ensemble system can leverage the benefits of each approach and achieve higher accuracy.

Also, making mBERT's decision-making process more explainable and interpretable through techniques like attention visualization or rule extraction can help increase transparency and trust. Researching cross-lingual transfer learning, where knowledge is transferred from high-resource

languages to low-resource languages, can help overcome data paucity challenges. Finally, having mechanisms for identifying and tracking false news on social media and news portals in real time is crucial in order to contain the rapid spread of misinformation. Through these lines of research, we can aid in the establishment of more robust, reliable, and responsive systems for detecting fake news that ultimately result in a more informed and credible information ecosystem.

Acknowledgments

The authors are grateful to Department of Electronics and Computer Engineering, Thapathali Campus for the provision of research facilities.

References

- [1] Bhuwan Bhandari. Weaponizing information: The rise of social media manipulation in nepal. *Journal of Development and Leadership*, 3(1), 2023. Accessed: 22 March 2025.
- [2] Ujjwal Acharya and Center for Media Research Nepal. Fact checking in nepal: How misinformation challenges mass media authenticity. *ResearchGate*, 2025. Citations: 0, Reads: 20, Accessed: 22 March 2025.
- [3] A. De et al. A Transformer-Based Approach to Multilingual Fake News Detection in Low-Resource Languages. *ACM Transactions on Asian Low Resource Language Information Processing*, 21(4):1–20, 2022.
- [4] N. N. Prachi et al. Detection of Fake News Using Machine Learning and Natural Language Processing Algorithms. *Journal of Advances in Information Technology*, 13(6):652–661, 2022.
- [5] M. Al-Yahya et al. Arabic Fake News Detection: Comparative Study of Neural Networks and Transformer-Based Approaches. *Complexity*, 2021:1–15, 2021.
- [6] A. M. S. Sánchez Gálvez et al. A study on the Detection of Fake news in Spanish. *International Journal of Combinatorial Optimization Problems and Informatics*, 15(2):1–10, 2024.
- [7] Sheetal Sharma, Pragyan Joshi, and Edwin van Teijlingen. 'nepenglish' or 'nepali english': A new version of english? *Asian Journal of Social Science and Humanities*, 4(3):188, 2015. Accessed: 22 March 2025.
- [8] NapoleonCat. Social media users in nepal – february 2024, 2024. Accessed: 22 March 2025.
- [9] Center for Media Research – Nepal. 95Accessed: 22 March 2025.
- [10] Djamila Romaina Beddiar, Md Saroar Jahan, and Mourad Oussalah. Data expansion using back translation and paraphrasing for hate speech detection. *Under Review*, 2024. Email: Djamila.Beddiar@oulu.fi.
- [11] A. Hande et al. Evaluating Pretrained Transformer-based Models for COVID-19 Fake News Detection. In *International Conference on Computing Methodologies and Communication*, 2021.
- [12] S. F. N. Azizah et al. Performance Analysis of Transformer Based Models (BERT, ALBERT, and RoBERTa) in Fake News Detection. In *2023 6th International Conference on Information and Communications Technology (ICOIACT)*, 2023.

- [13] A. Glazkova. Data Augmentation for Fake News Detection by Combining Seq2seq and NLI. In *Recent Advances in Natural Language Processing*, 2023.
- [14] Neha Shakya and Prakash Poudyal. Detection of fake news using deep neural networks. *Kathmandu University Journal of Science, Engineering and Technology*, 16(2):1–10, December 2022.
- [15] M. Siino et al. Fake news spreaders detection: Sometimes attention is not all you need. *Information*, 13(9):1–16, 2022.
- [16] Rakshya Sharma, Bidhya Sharma, Prashikshya Baral, and Susmita Parajuli. Nepali fake news detection using machine learning algorithms. *International Journal of Scientific Research and Engineering Development*, 7, July-Aug 2024.